

Hidden layer activator: ReLU
 Output activator: Softmax
 Output layer: 10 neurons
 Input layer: $(784 \times N_p) \rightarrow 28 \times 28$ image
 Loss function: Cross-Entropy loss

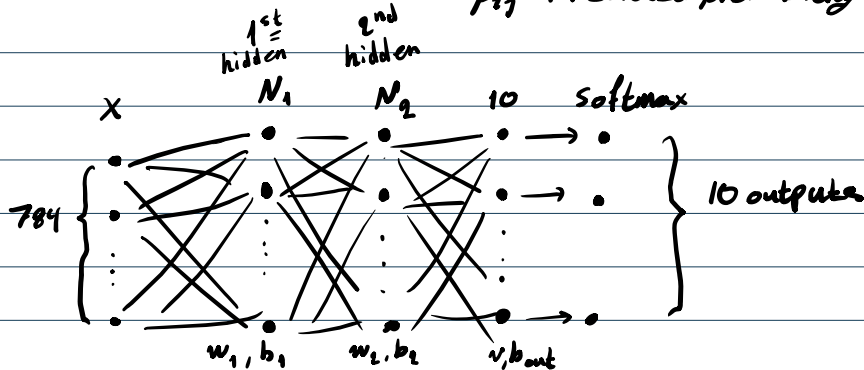
} NN architecture

$$L = -\frac{1}{N_p} \sum_{i=1}^{N_p} \sum_{j=1}^{N_c} y_{ij} \log(p_{ij})$$

N_p : Number of samples
 N_c : Number of classes

y_{ij} : "1" if sample i is of class j

p_{ij} : Predicted probability of sample i being of class j



$$\begin{aligned}
 X &\rightarrow (784 \times N_p) & w_2 &\rightarrow (N_2 \times N_1) & b_2 &\rightarrow (N_2 \times 1) & v &\rightarrow (10 \times N_2) \\
 \hat{y} &\rightarrow (10, N_p) & w_1 &\rightarrow (N_1 \times 784) & b_1 &\rightarrow (N_1 \times 1) & b_{out} &\rightarrow (10 \times 1)
 \end{aligned}$$

Forward path:

$$\left. \begin{aligned}
 \hat{y} &= \sigma_{sm}(vA_2 + b_{out}) \\
 A_2 &= \sigma_{ReLU}(w_2A_1 + b_2) \\
 A_1 &= \sigma_{ReLU}(w_1X + b_1)
 \end{aligned} \right\} \Rightarrow \hat{y} = \sigma_{sm}\left(v\sigma_{ReLU}(w_2\sigma_{ReLU}(w_1X + b_1) + b_2) + b_{out}\right)$$

Loss function in matrix form: $L = -\frac{1}{N_p} \text{tr}(y^T \log \hat{y})$

$$\text{ReLU}(z) = \begin{cases} x, & x \geq 0 \\ 0, & x < 0 \end{cases}, \quad \text{Softmax}(z) = \frac{e^z}{\sum_i e^{z_i}} = \frac{e^z}{\mathbf{1}^T \mathbf{e}^z}$$

$$\hat{y} = \frac{e^{vA_2 + b_{out}}}{\mathbf{1}^T e^{vA_2 + b_{out}}} \Rightarrow \log \hat{y} = (vA_2 + b_{out}) - \log(\mathbf{1}^T \underbrace{e^{vA_2 + b_{out}}}_{\mathbf{z}_{out}}) = \mathbf{z}_{out} - \log(\mathbf{1}^T \mathbf{e}^{\mathbf{z}_{out}})$$

$$dL = -\frac{1}{N_p} \text{tr}[d(y^T \log \hat{y})] = -\frac{1}{N_p} \text{tr}(y^T d \log \hat{y})$$

$$dL = -\frac{1}{N_p} \text{tr} [d(y^T \log \hat{y})] = -\frac{1}{N_p} \text{tr} (y^T d \log \hat{y})$$

$$d \log \hat{y} = dz_{out} - d \log(\bar{I}^T e^{z_{out}}) = dz_{out} - \frac{\bar{I}^T e^{z_{out}} dz_{out}}{\bar{I}^T e^{z_{out}}} = dz_{out} - \frac{(e^{z_{out}})^T \bar{I} dz_{out}}{\bar{I}^T e^{z_{out}}} =$$

$$dz_{out} - \frac{(e^{z_{out}})^T \bar{I} dz_{out}}{\bar{I}^T e^{z_{out}}} = (I - \bar{I} \hat{y}^T) dz_{out}$$

$$dL = -\frac{1}{N_p} \text{tr} \left[\underbrace{(y^T - y^T \bar{I} \hat{y}^T)}_1 dz_{out} \right] \Rightarrow dL = \frac{1}{N_p} \text{tr} \left[\underbrace{(\hat{y} - y)}_{\varepsilon} dz_{out} \right] = \frac{1}{N_p} \text{tr} (\varepsilon^T dz_{out})$$

$$\text{For } \nabla_v L: dz_{out} = d(vA_2 + b_{out}) = (dv)A_2, \quad dL = \frac{1}{N_p} \text{tr} [\varepsilon^T (dv)A_2] = \frac{1}{N_p} \text{tr} (A_2 \varepsilon^T dv) \Rightarrow \nabla_v L = \frac{1}{N_p} \varepsilon A_2^T$$

$$\text{For } \nabla_{b_{out}} L: dz_{out} = d(vA_2 + b_{out}) = db_{out}, \quad dL = \frac{1}{N_p} \text{tr} (\varepsilon^T db_{out}) \Rightarrow \nabla_{b_{out}} L = \frac{1}{N_p} \varepsilon$$

$$\text{For } \nabla_{w_2} L: dz_{out} = d(vA_2 + b_{out}) = v dA_2 = v d\sigma_{ReLU}(w_2 A_1 + b_2) = v \sigma'_{ReLU} \odot dw_2 A_1$$

$$dL = \frac{1}{N_p} \text{tr} (\varepsilon^T v \sigma'_{ReLU} \odot dw_2 A_1) = \frac{1}{N_p} \text{tr} ((v^T \varepsilon \odot \sigma'_{ReLU})^T dw_2 A_1) = \frac{1}{N_p} \text{tr} (A_1 (v^T \varepsilon \odot \sigma'_{ReLU})^T dw_2)$$

$$\nabla_{w_2} L = \frac{1}{N_p} \underbrace{(v^T \varepsilon \odot \sigma'_{ReLU})}_{\varepsilon_{h_2}} A_1^T = \frac{1}{N_p} \varepsilon_{h_2} A_1^T$$

$$\text{For } \nabla_{b_2} L: dz_{out} = v dA_2 = v d\sigma_{ReLU}(w_2 A_1 + b_2) = v \sigma'_{ReLU} \odot db_2, \quad dL = \frac{1}{N_p} \text{tr} (\varepsilon^T v \sigma'_{ReLU} \odot db_2)$$

$$= \frac{1}{N_p} \text{tr} [(\varepsilon^T v)^T \odot \sigma'_{ReLU}]^T db_2] \Rightarrow \nabla_{b_2} L = \frac{1}{N_p} (\varepsilon^T v)^T \odot \sigma'_{ReLU} = \frac{1}{N_p} \varepsilon_{h_2}$$

$$\text{For } \nabla_{w_1} L: dz_{out} = v dA_2 = v d\sigma_{ReLU}(w_2 A_1 + b_2) = v (\sigma'_{ReLU} \odot w_2 dA_1) = v (\sigma'_{ReLU} \odot w_2 d\sigma_{ReLU}(w_1 X + b_1))$$

$$= v (\sigma'_{ReLU} \odot (w_2 (\sigma'_{ReLU} \odot dw_1 X))), \quad dL = \frac{1}{N_p} \text{tr} (\varepsilon^T v (\sigma'_{ReLU} \odot (w_2 (\sigma'_{ReLU} \odot dw_1 X))))$$

$$= \frac{1}{N_p} \text{tr} \left(\underbrace{(v^T \varepsilon \odot \sigma'_{ReLU})}_{\varepsilon_{h_2}} w_2 (\sigma'_{ReLU} \odot dw_1 X) \right) = \frac{1}{N_p} \text{tr} \left((w_2^T \varepsilon_{h_2} \odot \sigma'_{ReLU})^T dw_1 X \right) =$$

$$\frac{1}{N_p} \text{tr} (X (w_2^T \varepsilon_{h_2} \odot \sigma'_{ReLU})^T dw_1) \Rightarrow \nabla_{w_1} L = \frac{1}{N_p} \underbrace{(w_2^T \varepsilon_{h_2} \odot \sigma'_{ReLU})}_{\varepsilon_{h_1}} X^T = \frac{1}{N_p} \varepsilon_{h_1} X^T$$

$$\text{For } \nabla_{b_1} L: dz_{out} = v (\sigma'_{ReLU} \odot (w_2 (\sigma'_{ReLU} \odot db_1))), \quad dL = \frac{1}{N_p} \text{tr} (\varepsilon v (\sigma'_{ReLU} \odot (w_2 (\sigma'_{ReLU} \odot db_1))))$$

$$= \frac{1}{N_p} \text{tr} ((v^T \varepsilon \odot \sigma'_{ReLU})^T w_2 (\sigma'_{ReLU} \odot db_1)) = \frac{1}{N_p} ((w_2^T \varepsilon_{h_2} \odot \sigma'_{ReLU})^T db_1) \Rightarrow \nabla_{b_1} L = \frac{1}{N_p} w_2^T \varepsilon_{h_2} \odot \sigma'_{ReLU} = \frac{1}{N_p} \varepsilon_{h_1}$$

$$= \frac{1}{N_p} \text{tr} \left((v^T \varepsilon \odot \sigma'_{\text{relu}})^T w_i (\sigma'_{\text{relu}} \odot db) \right) = \frac{1}{N_p} \left((w_i^T \varepsilon_{h_2} \odot \sigma'_{\text{relu}})^T db \right) \Rightarrow \nabla_{b_i} \mathcal{L} = \frac{1}{N_p} w_i^T \varepsilon_{h_2} \odot \sigma'_{\text{relu}} = \frac{1}{N_p} \varepsilon_{h_2}$$